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CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 03 – General Proficiency

1 hour

08 JUNE 2006 (a.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

In addition to the 1 hour, candidates are allowed a reading time of 10 minutes. Writing may begin during the 10-minute period.

This paper is divided into THREE sections. Answer ONE question from EACH section. Where appropriate, answers should be illustrated by diagrams.

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SECTION A

Answer ONE question from this section.

1. (a) With the aid of a diagram, show how the nephron (uriniferous tubule) is suited to its functions. **(8 marks)**

(b) An individual with malfunctioning kidneys is often required to undergo dialysis. In this process the blood is made to flow alongside a dialysis solution, which sets up a gradient with the blood to filter its impurities.

(i) What substances should be present in the dialysis solution? **(2 marks)**

(ii) In what ways would the dialysis process be **different** from the process that takes place in a properly functioning uriniferous tubule? **(6 marks)**

(c) (i) Give TWO ways in which plants eliminate waste products from their systems. **(2 marks)**

(ii) Suggest why the mechanisms plants use to eliminate waste products are not as complex as those of animals. **(2 marks)**

Total 20 marks

2. (a) With the aid of a labelled diagram, explain how the human eye is well suited to its functions. **(8 marks)**

(b) Eyeglasses (spectacles) are used to replace or supplement the functioning of the lens.

(i) Explain how eyeglasses are able to correct for images falling behind and images falling in front of the retina. **(4 marks)**

(ii) How do the lenses of eyeglasses function differently from properly functioning lenses of the eyes? **(4 marks)**

(c) Contact lenses function similarly to eyeglasses but are worn directly on the surface of the eye.

Suggest ONE advantage and ONE disadvantage of using contact lenses. **(4 marks)**

Total 20 marks

SECTION B

Answer ONE question from this section.

3. (a) Living organisms experience a life cycle even though each organism lives for a consecutive number of years.

(i) Using a named example, explain why the stages in the life of an organism are described as a cycle. **(4 marks)**

(ii) Draw a labelled representation of the stages in the life of the organism you named at (a) (i). **(4 marks)**

(b) (i) With the aid of diagrams, describe the process of mitosis. **(8 marks)**

(ii) Explain the importance of mitosis in the life of a named organism. **(4 marks)**

Total 20 marks

4. (a) (i) Explain why it is important for an animal to have organs of sensitivity. **(2 marks)**

(ii) Describe how the skin is equipped to carry out its **sensory** function. **(2 marks)**

(iii) Draw a simple representation of the path impulses would follow if the sense organ referred to in (a) (ii) were part of a reflex arc. **(4 marks)**

(b) Plants do not usually possess elaborate sense organs, yet there is often the need for them to perform special functions. Two such special functions are pollination and fertilization.

(i) Explain why plants depend on external factors or conditions to help in the pollination process. **(2 marks)**

(ii) Suggest FOUR ways in which a plant may be adapted for a named method of pollination. (Indicate the method of pollination for suggested adaptation). **(4 marks)**

(iii) With the aid of a diagram, describe how pollination leads to fertilization. **(6 marks)**

Total 20 marks

SECTION C

Answer ONE question from this section.

5. (a) (i) For a named insect vector, state the habitat of EACH stage in its life cycle. **(4 marks)**

(ii) Name TWO diseases spread by the activity of a vector **and** for any ONE disease explain the role of the vector. **(4 marks)**

(b) A particular Caribbean village is plagued by an insect vector. Some villagers suggest spraying to control the insects while others do not agree with that method.

(i) Why do some of the villagers choose NOT to support the use of sprays to control the insects? **(4 marks)**

(ii) Identify TWO methods OTHER THAN spraying that could be used to eradicate the insect vector you named at (a) (i) above **and** explain why these methods would be preferred by some villagers. **(8 marks)**

Total 20 marks

6. (a) Using examples, distinguish between physical and biotic factors in the environment. **(4 marks)**

(b) In one Caribbean territory a family imports and privately keeps a few deer. The deer escape into nearby forests during a hurricane and their population quickly increases to a few hundreds. Deer are herbivores and are not part of the natural fauna (animal life) of that territory.

(i) Identify TWO biotic factors that could have helped the deer population to increase, and explain how EACH would have contributed to the increase. **(4 marks)**

(ii) How would the increase in the deer population affect the environment? **(4 marks)**

(c) Besides introducing new species, human activities also affect the natural environment in other ways. For example, many Caribbean coral reefs have been damaged by human activities.

(i) Describe TWO ways in which human activity destroys coral reefs. **(2 marks)**

(ii) Suggest the implications of the coral reef destruction described in (c) (i). **(6 marks)**

Total 20 marks

END OF TEST